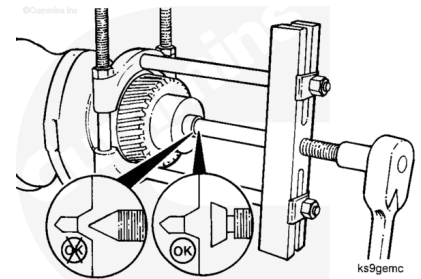


Trainings ▾

Remove

⚠ CAUTION ⚠

Do not try to split the front crankshaft gear to remove. The gear is made out of steel and will not split. Damage to the tool and crankshaft can result.



Note : If a front crankshaft seal wear sleeve has been installed during a previous repair, it **must** be removed before removing the crankshaft gear. Refer to Procedure 001-025 (</qs3/pubsys2/xml/en/procedures/40/40-001-025.html>).

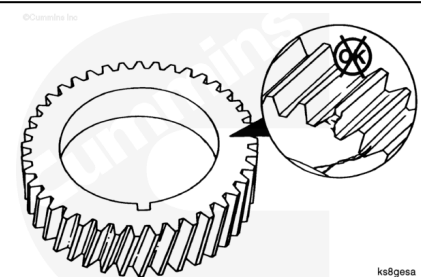
Use a heavy-duty bearing separator, Part Number 3162427, or a gear puller, as illustrated, to remove the crankshaft gear.

Clean and Inspect for Reuse

Visually inspect for cracks and broken or chipped teeth.

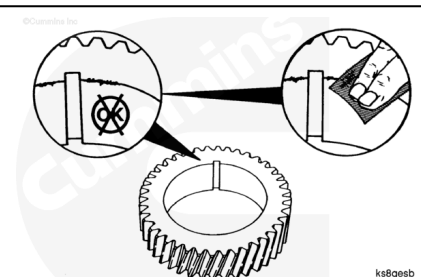
The gear **must** be replaced if it is damaged.

Note : If there is damage to the front crankshaft gear teeth or there are signs of excessive heat, make sure to inspect the associated lubricating oil pump and, for front gear train engines, the fuel pump and camshaft gears for damage.



Visually inspect the gear and, for front gear train engines only, the keyway for nicks or burrs.

Use fine crocus cloth to remove nicks and burrs.

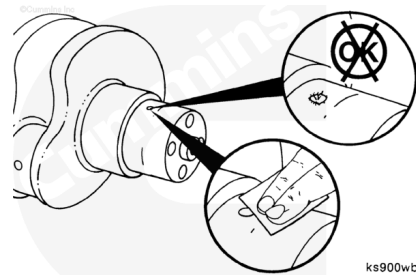


Visually inspect the crankshaft gear journal, gear alignment dowel pin and dowel pin hole for burrs or damage.

Use fine crocus cloth to remove burrs.

Replace the gear alignment dowel pin if damage is found.

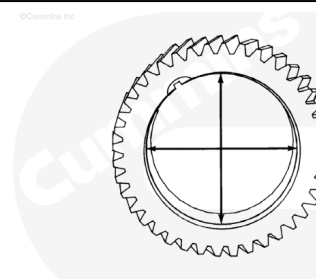
Note : Rear gear train engines do not have any type of alignment pin for the front crankshaft gear.



ks900wb

Measure the crankshaft gear bore inside diameter.

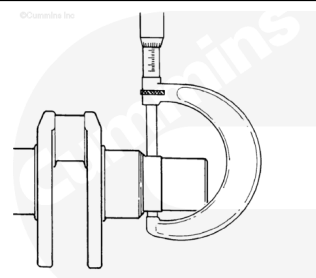
Crankshaft Gear Bore Inside Diameter		
mm		in
63.910	MIN	2.516
63.934	MAX	2.517



ks1geta

Measure the crankshaft gear journal outside diameter.

Crankshaft Gear Journal Outside Diameter		
mm		in
63.475	MIN	2.499
63.525	MAX	2.500



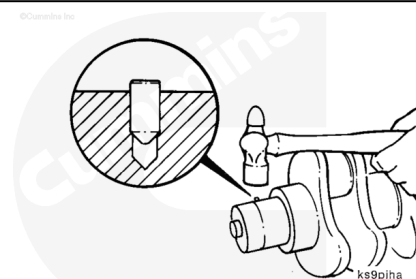
ks1getb

Install

Use a soft hammer to install the gear alignment dowel into the crankshaft.

The pin **must** be 1.02 mm [0.040 in] above the crankshaft surface.

Note : Rear gear train engines do not have any type of alignment pin for the front crankshaft gear.

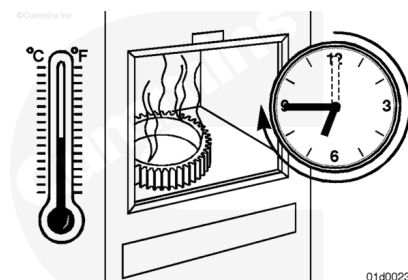


ks9piha

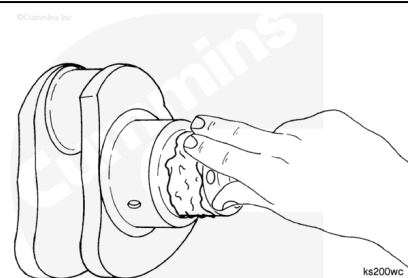
⚠ CAUTION ⚠

Do not exceed the specified heating time or temperature. The crankshaft teeth can be damaged.

Heat the gear in an oven for a minimum of 45 minutes, but **not** more than 2 hours at 177°C [350°F].



Use assembly lube, Part Number 3163087 or equivalent, to lubricate the outside diameter of the crankshaft gear journal.

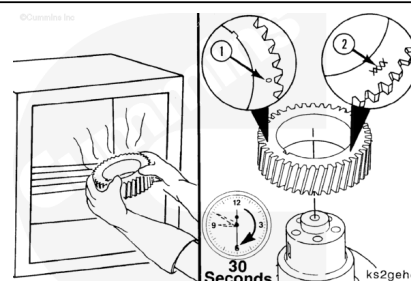


⚠ WARNING ⚠

Wear protective gloves to reduce the possibility of personal injury when handling parts that have been heated.

⚠ CAUTION ⚠

For front gear train engines the timing mark (1) and part number (2) on the gear must be facing away from the crankshaft after the gear is installed. Engine damage can result if the gear is installed backwards.



⚠ CAUTION ⚠

Do not use water or oil to reduce the cooling time. The gear can crack. Allow the gear to air cool.

Note : Rear gear train engines do not have any type of keyway or timing mark on the front crankshaft gear. The gear can be installed in any orientation.

Remove the gear from the oven.

Align the keyway of the gear with the alignment dowel pin in the crankshaft and install the gear within 30 seconds.

For front gear train engines, align the keyway of the gear with the alignment dowel pin in the crankshaft. Install the crankshaft gear within 30 seconds of removing it from the oven.

If the gear cools and stops on the crankshaft before it is fully installed, use a driver to complete the installation.

Make sure the gear is seated against the crankshaft shoulder. Use a 0.02 mm [0.001 in] feeler gauge to check to see if the feeler gauge can be inserted between the crankshaft gear and the shoulder on the crankshaft. If the feeler gauge can be inserted, the crankshaft gear is **not** properly seated and **must** be removed and installed again.

